

Solve a System of Equations

MATLAB matrix solution and Command Window output. The screenshot reports $V_A = 4.218750$ V, $V_B = 0.781250$ V, and $V_C = 0.781250$ V.

```
/MATLAB Drive/Activity_03_solve_system.m
1 %% Proof of Skills - Solve a System of Equations
2 % Unknown vector x is [VA; VB; VC]. The equations are:
3 % VA/100 = VB/20 + VC/250
4 % VA + VB = 5
5 % -VB + VC = 0
6
7 clear;
8 clc;
9
10 A = [1/100, -1/20, -1/250; ...
11      1,      1,      0; ...
12      0,      -1,     1];
13 b = [0; 5; 0];
14
15 x = A \ b;
16 VA = x(1);
17 VB = x(2);
18 VC = x(3);
19 residual = A*x - b;
20
21 disp('Coefficient matrix A:');
22 disp(A);
23 disp('Right-hand side b:');
24 disp(b);
25 fprintf('VA = %.6f V\n', VA);
26 fprintf('VB = %.6f V\n', VB);
27 fprintf('VC = %.6f V\n', VC);
28 fprintf('Maximum absolute residual = %.3e\n', max(abs(residual)));
29
```

Command Window

New to MATLAB? See resources for [Getting Started](#).

```
Coefficient matrix A:
0.0100    -0.0500   -0.0040
1.0000    1.0000    0
0         -1.0000    1.0000

Right-hand side b:
0
5
0

VA = 4.218750 V
VB = 0.781250 V
VC = 0.781250 V
Maximum absolute residual = 2.602e-18
>>
```

MATLAB code and output for the linear system.